

Question 9: What would be the output of the following code?

```
import abc

class BankAccount(abc.ABC):
    def __init__(self, account_number, account_holder, initial_deposit):
        self.account_number = account_number
        self.account_holder = account_holder
        self.balance = 0
        self.deposit(initial_deposit)

    @abc.abstractmethod
    def deposit(self, amount):
        pass

    @abc.abstractmethod
    def withdraw(self, amount):
        pass

class SavingsAccount(BankAccount):
    def deposit(self, amount):
        if amount > 0:
            self.balance += amount

    def withdraw(self, amount):
        deduction = 0.05 * amount
        self.balance -= (amount + deduction)

if __name__ == "__main__":
    my_account = BankAccount("92338723", "John Smith", 5000)
    my_account.withdraw(500)
    print(my_account.balance)
```

- A. It won't print since the withdraw method is an abstract method
- B. 4475
- C. The code will throw an Error since it can't instantiate an object with abstract methods
- D. 0

Correct Answer: C. The code will throw an Error since it can't instantiate an object with abstract methods

Question 10: Select all true statements about Abstract Base Classes

- A. An Abstract Base Class defines an interface for child classes
- B. Abstract methods use @abc.abstractmethod
- C. You can instantiate an Abstract Base Class
- D. Child classes must override abstract methods to be instantiated

Correct Answer: A, B, D

Question 11: What direction should the dependency arrow face?

```
class Book:
    def __init__(self, title, author):
        self._title = title
        self._author = author

class Printer:
    def print_book_details(self, book: Book):
        print(book.get_title())
```

- A. Printer <--> Book
- B. Printer -> Book
- C. Printer <- Book
- D. No dependency

Correct Answer: B. Printer -> Book

Question 12: Which is a type of formal interface in Python?

- A. Abstract base classes
- B. Duck typing
- C. Protocols

Correct Answer: A. Abstract base classes

Question 13: Coupling is not a type of Dependency.

- A. True
- B. False

Correct Answer: B. False